

Course Code : ECT 357

QTSR/RW – 24 / 1665

**Sixth Semester B. Tech.(Electronics and Communication Engineering)
Examination**

COMPUTER NETWORKS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. Explain role of each layer in ISO-OSI reference Model, with suitable diagrams. 10(CO1)
2.
 - (a) Justify the statement ‘Packet switching is mainly used for data communication in the internetworking’. Also, explain various types of packet switching methods. 5(CO1,2)
 - (b) In the layered protocol suits various network connecting devices are used at various layers. Name the network connecting devices used at Data Link layer and Network layers. Explain their uses at particular layer. 5(CO2)
3.
 - (a) Generate a CRC for the data $(238)_2$ using the divisor 1101. Also explain the procedure to detect the error, if any in the received CRC at the receiver. 5(CO4)
 - (b) Justify the statement - “collision is an issue in a random access protocol but not in controlled access protocol” 5(CO2)
4.
 - (a) Illustrate various fields of IPv4 and IPv6 headers. 5(CO5)

- (b) Predict if the two devices given below are on the same subnet of different subnets. The address/mask combinations, written with the prefix/length notation are-
- Device A : 172. 16. 17. 30 / 20
- Device B : 172. 16. 28. 15 / 20 5(CO5)
5. (a) Enlist various congestion control protocols used at Transport layer. Explain any one in brief. 5(CO4)
- (b) Differentiate between congestion control and flow control. Briefly discuss various types of congestion control algorithms. 5(CO4)
6. (a) Illustrate digital signature used in cryptography techniques. 5(CO3)
- (b) Explain how the concept of symmetric key cipher is Implemented in computer network security. 5(CO3)

